

It's not you, it's the math:
your friends are (probably) more important than you

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ABSTRACT

This study provides a comprehensive empirical and theoretical investigation of the Generalized Friendship Paradox across diverse network topologies. By evaluating global and local friendship gaps across six standard centrality metrics, we show that real-world networks have very unequal structures, with a few highly connected individuals. This high inequality strongly increases the illusion that our friends are more popular or important than we are, far more than in purely Erdős–Rényi random networks models. Furthermore, we introduce a novel localized centrality metric, mapping a node’s relative standing within its immediate neighborhood and offering a complementary perspective to traditional global centrality measures.

Github Repository: <https://github.com/samueltia/generalized-friendship-paradox>

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List of Symbols

Symbol	Description
$G = (V, E)$	Simple, undirected graph with vertex set V and edge set E
$n = V $	Total number of vertices (nodes) in the graph
$m = E $	Total number of edges in the graph
d_i	Degree of node i , i.e., $d_i = N(i) = \sum_j A_{i,j}$
$N(i)$	Open neighborhood of node i , i.e., $\{j \in V : (i, j) \in E\}$
$\bar{k} = \langle k \rangle$	Average degree of the network, $\frac{1}{n} \sum_{i=1}^n d_i = \frac{2m}{n}$
$\mathbf{A}$	Adjacency matrix of the graph ($A_{i,j} = 1$ if $(i, j) \in E$, 0 otherwise)
$\mathbf{D}$	Diagonal degree matrix, $\text{diag}(d_1, d_2, \dots, d_n)$
$\mathbf{x}$	Generic vector of node importances or centrality scores ($\mathbf{x} \in \mathbb{R}^n$)
$\mathbf{1}$	Column vector of all ones ($\mathbf{1} \in \mathbb{R}^n$)
$\lambda_{\max}$	Principal eigenvalue of the adjacency matrix $\mathbf{A}$
α	Attenuation/scaling parameter for Katz and Exponential centralities
$\mathbf{B}$	Non-backtracking matrix used for non-backtracking centrality
p	Edge creation probability parameter in the Erdős–Rényi $G(n, p)$ model
$\mathbb{E}[x]$	Expected centrality of a node chosen uniformly at random ($\frac{1}{n} \sum x_i$)
$\mathbb{E}[x_{\text{neigh}}]$	Expected centrality of a randomly selected neighbor ($\frac{1}{2m} \sum d_i x_i$)
Δ_{GFP}	Global friendship paradox gap, $\mathbb{E}[x_{\text{neigh}}] - \mathbb{E}[x]$
Δ_i	Local friendship gap for node i , i.e., $\frac{1}{d_i} \sum_j A_{i,j} x_j - x_i$
Δ	Vector of local friendship gaps across all nodes, $\mathbf{D}^{-1} \mathbf{A} \mathbf{x} - \mathbf{x}$
$C_{\text{local}}(i)$	Exponential local centrality metric defined as $\exp(-\Delta_i)$
s_i	Local score scaling parameter for node i ($s_i = -\Delta_i$)
$\hat{\mu}, \hat{\sigma}$	Maximum Likelihood Estimates (MLE) for mean and standard deviation
$f(\Delta_i; \hat{\mu}, \hat{\sigma})$	Gaussian probability density function fitted to local gaps
γ_1	Sample skewness quantifying distributional asymmetry
$F_N(x)$	Empirical cumulative distribution function (ECDF) over active nodes
$\Phi(z)$	Cumulative distribution function (CDF) of the standard normal distribution
D	Kolmogorov–Smirnov (KS) distance statistic relative to normal fit
$r(u, v)$	Pearson correlation coefficient between node metric vectors u and v
$\text{Corr}(\exp(-\Delta_i), x)$	Pearson correlation between local dominance and global centrality x

List of Abbreviations

Abbreviation	Full Form
FP	Friendship Paradox
GFP	Generalized Friendship Paradox
LFP	Local Friendship Paradox
ER	Erdős–Rényi (random graph model)
BA	Barabási–Albert (scale-free graph model)
WS	Watts–Strogatz (small-world graph model)
GNN	Graph Neural Network
MLE	Maximum Likelihood Estimation
ECDF	Empirical Cumulative Distribution Function
CDF	Cumulative Distribution Function
KS	Kolmogorov–Smirnov (statistical test / distance metric)

1 Introduction

1.1 Motivation and Context

The Friendship Paradox (FP), first articulated by Feld in 1991, states that "on average, your friends have more friends than you do" [1]. This counterintuitive phenomenon arises from a fundamental sampling bias in networks: when individuals are sampled through their connections, highly connected nodes are more likely to be encountered than nodes with fewer connections. As a result, the local view of a network can differ substantially from its global structure.

The Generalized Friendship Paradox (GFP) extends this idea beyond degree by considering other measures of node importance, including centrality measures and node attributes [2]. It shows that similar biases can arise across different dimensions of network structure, suggesting that the way a network is sampled can strongly influence how the importance of its nodes is perceived.

1.2 Practical Applications of the Friendship Paradox

While initially a curiosity in social psychology, the FP has since been recognized as a critical factor affecting diverse domains because highly connected individuals are more likely to appear when we observe the friends of randomly selected individuals.

- **Public health and epidemic control.** Monitoring the friends of randomly selected individuals can help identify highly connected people who may play an important role in disease transmission [3]. This idea can be used for early outbreak detection and for designing more efficient vaccination or monitoring strategies.
- **Information detection and diffusion.** Highly connected individuals are more likely to encounter new information early and to spread it quickly through a network [4]. Observing the friends of a random sample can therefore help detect emerging trends, products, or opinions, while targeting well-connected individuals can facilitate the rapid dissemination of important information.
- **Social media and social comparison.** The paradox also helps explain why people on social media may appear more popular or socially active than they really are [5]. Highly connected users are naturally overrepresented in other users' networks and feeds.

1.3 Research Questions

To evaluate how local perceptions differ from global network structures, we quantify the local friendship gap Δ_i , defined as the difference between the average importance of a node's immediate neighbors and its own importance:

The local friendship gap Δ_i

$$\Delta_i = \frac{1}{d_i} \sum_j A_{i,j} x_j - x_i \quad (1)$$

where $A_{i,j}$ represents the elements of the adjacency matrix, d_i is the degree of node i , and $\mathbf{x}$ is the vector of node importances. Building on this framework, this work addresses the following research questions:

- **RQ1:** How does the Global Friendship Paradox behave across different node centrality measures, and how do real-world graph structures deviate from homogeneous random networks (specifically Erdős–Rényi models) where nodes lack privileged positions?

- **RQ2:** How does the Local Friendship Paradox vary across real-world and random networks, and what structural properties of a network are associated with the local gap Δ_i ?
- **RQ3:** Can $\exp(-\Delta_i)$ provide a meaningful centrality measure, and how does it compare with existing centrality measures?

1.4 Research Contributions and Structure of the Work

This work provides a systematic empirical and theoretical analysis of both global and local formulations of the Friendship Paradox across real-world networks and synthetic null models. The main contributions of this paper are summarized as follows:

1. **Comprehensive Evaluation of the Global Friendship Paradox:** We benchmark the Global Friendship Paradox across six distinct centrality measures (Degree, Eigenvector, Walk, Katz, Exponential, and Non-backtracking centralities) on five diverse empirical networks and their corresponding Erdős–Rényi random graphs.
2. **Characterization of the Local Friendship Gap (Δ_i):** We formalize the local friendship gap Δ_i for general importance vectors x , analyzing its empirical distribution, expected value, and proportion of paradoxical nodes ($\Delta_i \geq 0$) across networks.
3. **Analytical Random Graph Derivation:** We establish theoretical normal approximations for the distribution of Δ_i in Erdős–Rényi random graphs and validate them empirically.
4. **Novel Centrality Metric $\exp(-\Delta_i)$:** We introduce $\exp(-\Delta_i)$ as a localized centrality measure reflecting relative node standing within immediate neighborhoods and analyze its correlation with traditional global centralities.

The remainder of this report is organized as follows: Section 2 presents on the friendship paradox. Section 3 provides the mathematical foundations and graph notation. Section 4 details the datasets and experimental methodology. Section 5 presents the global framework of the Friendship Paradox. Section 6 analyzes the local gap Δ_i . Section 7 evaluates the proposed centrality metric $\exp(-\Delta_i)$, and Section ?? concludes the work. Additional figures and empirical distributions are compiled in the Appendix.

2 Related Work

The Friendship Paradox, introduced by Feld [1], established that an individual’s friends have, on average, more friends than the individual themselves due to sampling bias toward highly connected nodes. Over the past decade, this foundational concept has been significantly expanded to investigate whether similar biases persist for non-degree attributes, leading to the mathematical formulation of the Generalized Friendship Paradox [6, 2].

2.1 Empirical Foundations

Eom and Jo [6] were among the first to formalize the GFP for arbitrary node characteristics across real-world scientific collaboration networks. They demonstrated that the GFP holds at both individual and aggregate levels for non-degree characteristics such as citation counts, publication volume, and co-author numbers. Their work established that the mathematical origin of the GFP stems directly from positive correlations between node degree and target characteristics, and suggested leveraging the GFP for efficient sampling strategies to identify highly influential nodes in large networks.

2.2 Analytical Models

Extending beyond mean behavior, Cantwell et al. [7] developed a comprehensive mathematical framework for both the classical and generalized friendship paradoxes across random network models and empirical datasets. Using generating function methods, they derived the full probability distributions of neighbor degree differences, accounting for variance around expected values. Their empirical comparisons across a vast collection of real-world datasets confirmed strong agreement with theoretical null model predictions while emphasizing the role of network structure in driving local variance.

2.3 Linear Algebra and Centrality Extensions

Most recently, Higham et al. [2] derived new theoretical bounds for the GFP using a linear algebra perspective. Comparing global (degree-weighted) and local neighborhood averages, they proved that a global GFP holds for a broad family of walk-based centralities, such as Katz centrality and total subgraph communicability, as well as non-backtracking eigenvector centrality. However, they identified counterexamples for centralities based strictly on even-length walks. For local averaging, they characterized exact conditions where the local paradox holds and demonstrated that for specific inverse attributes (such as loneliness, defined as $1/d_i$), the paradox holds in reverse.

Building upon these foundational insights, our work systematically evaluates global and local GFP formulations across real-world networks and Erdős–Rényi synthetic graphs, while introducing $\exp(-\Delta_i)$ to capture localized neighborhood standing.

3 Mathematical Background

3.1 Graph Notation

Let us consider a graph with n nodes and m edges. We denote the adjacency matrix of this graph as A . Let $\mathbf{1} \in \mathbb{R}^n$ be the vector of all ones.

Graph Degree Matrix. The degree vector of the graph is defined as $d = A\mathbf{1}$, and $D = \text{diag}(d)$ represents the diagonal matrix containing the node degrees. Furthermore, let $x \in \mathbb{R}^n$ represent a generic vector of node importances or attributes.

3.2 Centrality Measures

To evaluate the structural importance of nodes within the network, several centrality measures can be defined based on the adjacency matrix A . Each metric captures a different perspective on what makes a node "central" or influential.

3.2.1 Degree Centrality

Degree Centrality

$$d = A\mathbf{1} \quad (2)$$

Intuition: This is the simplest measure of centrality. It merely counts the number of direct neighbors a node possesses. In a social network context, it represents immediate popularity or the direct reach a person has without relying on intermediaries.

3.2.2 Eigenvector Centrality

Eigenvector centrality assigns relative scores to all nodes in the network based on the principle that connections to high-scoring nodes contribute more to the score of the node in question.

Eigenvector Centrality

$$A\mathbf{x} = \lambda_{\max}\mathbf{x} \implies x_i = \frac{1}{\lambda_{\max}} \sum_{j=1}^n A_{ij}x_j \quad (3)$$

where $\mathbf{x}$ is normalized and guaranteed to have non-negative entries by the Perron–Frobenius theorem.

Intuition: Unlike degree centrality, which treats all connections equally, eigenvector centrality operates on the principle that connections to highly influential nodes contribute more to a node's own influence than connections to less influential nodes. It is a recursive measure where a node is important if its friends are also important.

3.2.3 Walk Centrality

Walk Centrality

$$\mathbf{x} = A^\ell \mathbf{1} \quad (4)$$

Intuition: Walk centrality counts the exact number of walks of length ℓ that originate

from a given node. It measures the ability of a node to spread or gather information exactly ℓ steps away, capturing a bounded sphere of influence.

3.2.4 Katz Centrality

Katz Centrality

Computed by solving the linear system:

$$(\mathbf{I} - \alpha \mathbf{A})\mathbf{x} = \mathbf{1} \implies \mathbf{x} = (\mathbf{I} - \alpha \mathbf{A})^{-1}\mathbf{1} = \sum_{k=0}^{\infty} \alpha^k \mathbf{A}^k \mathbf{1} \quad (5)$$

where $0 < \alpha < 1/\lambda_{\max}(\mathbf{A})$ is an attenuation parameter.

Intuition: Katz centrality extends the concept of walk centrality by counting walks of all possible lengths, applying an attenuation factor α to penalize longer walks. It grants a baseline centrality (the vector $\mathbf{1}$) to all nodes, meaning even a node with no incoming links maintains a minimal level of importance.

3.2.5 Total Subgraph Communicability (Exponential Centrality)

Exponential Centrality

Defined using the matrix exponential as:

$$\mathbf{x} = \exp(\alpha \mathbf{A})\mathbf{1} = \sum_{k=0}^{\infty} \frac{\alpha^k \mathbf{A}^k}{k!} \mathbf{1} \quad (6)$$

Intuition: Similar to Katz centrality, this measure considers walks of all lengths. However, it penalizes longer walks much more aggressively by dividing by the factorial of the walk length $k!$. It heavily emphasizes a node's participation in short, dense, local substructures.

3.2.6 Non-backtracking Eigenvector Centrality

Non-backtracking Eigenvector Centrality

Corresponds to the vector formed by taking the first n entries (normalized to have an L_2 -norm of 1) of the leading eigenvector $\mathbf{v}$ of the block matrix $\mathbf{B}$:

$$\begin{bmatrix} \mathbf{A} & \mathbf{I} - \mathbf{D} \\ \mathbf{I} & \mathbf{0} \end{bmatrix} \mathbf{v} = \lambda_{\max} \mathbf{v}, \quad \text{with } \mathbf{x} = \frac{\mathbf{v}_{1:n}}{\|\mathbf{v}_{1:n}\|_2} \quad (7)$$

Intuition: Standard walk-based centralities suffer from localization or "echo chamber" effects, where walks simply bounce back and forth between adjacent nodes, artificially inflating their importance. Non-backtracking centrality prevents a walk from immediately returning to the node it just left. This provides a more robust measure of influence, particularly in sparse networks or graphs with massive hubs.

3.3 Random Graphs: Erdős–Rényi

To study whether the structural patterns observed in real networks differ from purely random configurations, we utilize the Erdős–Rényi random graph model $G(n, p)$. In this work, we choose p such that the synthetic random graph has, in expectation, the exact same number of edges as the empirical target graph. The derivation of p proceeds as follows:

1. **Total Potential Edges:** In a simple undirected graph with n nodes, each pair of nodes can potentially form an edge. The total number of possible pairs is given by the binomial coefficient:

$$\binom{n}{2} = \frac{n(n-1)}{2} \quad (8)$$

In $G(n, p)$, each potential edge is present independently with probability p .

2. **Expected Number of Edges:** Let M denote the total number of edges in the random graph. We can express M as a sum of indicator random variables:

$$M = \sum_{1 \leq i < j \leq n} X_{ij} \quad (9)$$

where X_{ij} is defined as:

$$X_{ij} = \begin{cases} 1 & \text{if edge } (i, j) \text{ exists} \\ 0 & \text{otherwise} \end{cases} \quad (10)$$

Since $X_{ij} \sim \text{Bernoulli}(p)$, its expected value is $\mathbb{E}[X_{ij}] = p$. By the linearity of expectation, the expected total number of edges is:

$$\mathbb{E}[M] = \sum_{1 \leq i < j \leq n} \mathbb{E}[X_{ij}] = \binom{n}{2} p = \frac{n(n-1)}{2} p \quad (11)$$

3. **Matching the Real Graph’s Density:** Suppose our real-world graph possesses m edges. We require the expected number of edges in our random model to match this value, meaning $\mathbb{E}[M] = m$. Thus:

$$\frac{n(n-1)}{2} p = m \quad (12)$$

Isolating the probability p yields:

$$p = \frac{2m}{n(n-1)} \quad (13)$$

ER Model Density Matching : By setting $p = \frac{2m}{n(n-1)}$, the generated Erdős–Rényi random graph acts as a baseline preserving the exact overall density and edge count expectation of the empirical target graph.

4 Datasets and Experimental Methodology

4.1 Real-World Network Datasets

To empirically evaluate the behavior of the Friendship Paradox and our proposed local metrics across diverse topological structures, we select a representative collection of real-world networks from various domains. Table 1 summarizes the main topological properties of these empirical networks, including the number of nodes (n), number of edges (m), average degree ($\bar{k}$), density (p), and edge type (weighted or binary).

Table 1: Summary statistics of empirical real-world networks.

Dataset	Nodes (n)	Edges (m)	Avg. Degree ($\bar{k}$)	Density (p)	Edge Type
Karate Club [8]	34	78	4.59	0.1390	Weighted
Florentine Families [9]	15	20	2.67	0.1905	Binary
Les Miserables [10]	77	254	6.60	0.0868	Weighted
Cora [11]	2708	5278	3.90	0.0014	Binary
CiteSeer [12]	3327	4552	2.74	0.0008	Binary

Description of Topological Characteristics. As summarized in Table 1, these metrics allow us to compare the scale and global connectivity of each network:

- **Number of Nodes (n):** Represents the total size of the network by counting individual elements (e.g., individuals in a social network like *Karate Club*, or academic papers in *Cora* and *CiteSeer*).
- **Number of Edges (m):** Quantifies the total number of connections or interactions present between pairs of nodes (e.g., friendship ties, co-appearances, or citation links).
- **Average Degree ($\bar{k}$):** Measures the average number of connections per node, computed as $\bar{k} = \frac{2m}{n}$. As shown in Table 1, *Les Miserables* exhibits the highest average degree ($\bar{k} = 6.60$), reflecting tightly grouped interactions around central characters.
- **Density (p):** Reflects the proportion of actual edges relative to the maximum possible number of edges, calculated via $p = \frac{2m}{n(n-1)}$. Table 1 illustrates the clear contrast between small, highly dense social networks (such as *Florentine Families* with $p = 0.1905$) and large, extremely sparse citation networks (such as *CiteSeer* with $p = 0.0008$).
- **Edge Type:** Indicates whether edges carry a weight reflecting interaction intensity or co-occurrence frequency (*Karate Club* and *Les Miserables*), or are purely binary connections (*Florentine Families*, *Cora*, and *CiteSeer*). Note that n , m , $\bar{k}$, and p are computed on the unweighted (binary) structure in all cases, regardless of edge type, to keep these topological quantities comparable across datasets.

Dataset Structural Diversity : The dataset suite spans multiple orders of magnitude in both size ($n \in [15, 3327]$) and sparsity ($p \in [0.0008, 0.1905]$), allowing for a robust assessment of how local metrics scale from dense social cliques to sparse citation graphs.

4.2 Synthetic Network Generation

For each empirical dataset, we generate corresponding null models using the Erdős–Rényi random graph model $G(n, p)$.

Random Model Calibration. As derived in Section 3.3, the edge creation probability p is calibrated directly to preserve the expected density and total number of edges of the empirical target network:

$$p = \frac{2m}{n(n-1)} \quad (14)$$

Comparing real-world topologies with these matched Erdős–Rényi graphs allows us to isolate the specific effects of non-random network structures (such as heavy-tailed degree distributions and community structure) on the Friendship Paradox.

5 The Global Framework

5.1 Global Friendship Paradox

Using the generic vector of importances $\mathbf{x} \in \mathbb{R}^n$, the Global Friendship Paradox can be mathematically expressed as an inequality comparing the average centrality of a randomly chosen node to that of a randomly chosen neighbor across the network.

5.1.1 Mathematical Computation of Expected Values

Node-Level Expected Centrality

The expected centrality of a node chosen uniformly at random from the graph is given by the unweighted global mean:

$$\mathbb{E}[x] = \frac{\mathbf{1}^\top \mathbf{x}}{n} = \frac{1}{n} \sum_{i=1}^n x_i \quad (15)$$

Intuition: This measures the standard baseline average metric across the entire network, treating every node with equal probability ($1/n$).

Neighbor-Level Expected Centrality

When picking a random edge and following it to a node, the probability of selecting node i is proportional to its degree d_i . Thus, the expected centrality of a randomly chosen neighbor is represented by the degree-weighted average of the node importances:

$$\mathbb{E}[x_{\text{neigh}}] = \frac{\mathbf{d}^\top \mathbf{x}}{\mathbf{1}^\top \mathbf{d}} = \frac{1}{2m} \sum_{i=1}^n d_i x_i \quad (16)$$

Intuition: Sampling nodes via edges introduces a structural bias: highly connected hubs appear in many edge paths, making them much more likely to be sampled than peripheral nodes.

Global Friendship Paradox & Gap Definition. The Global Friendship Paradox states that the degree-weighted average centrality is greater than or equal to the unweighted global average:

$$\frac{\mathbf{d}^\top \mathbf{x}}{\mathbf{1}^\top \mathbf{d}} \geq \frac{\mathbf{1}^\top \mathbf{x}}{n} \quad (17)$$

The magnitude of this structural sampling bias is quantified by the global gap:

$$\Delta_{\text{GFP}} = \mathbb{E}[x_{\text{neigh}}] - \mathbb{E}[x] \quad (18)$$

5.2 Global Paradox Comparison in Real-world and Random Graphs

We examine the magnitude of Δ_{GFP} across empirical networks and their Erdős–Rényi baseline counterparts $G(n, p)$.

Figure 1 provides a visual comparison of individual mean centrality $\mathbb{E}[x]$ versus neighbor mean centrality $\mathbb{E}[x_{\text{neigh}}]$ across all empirical datasets and centralities.

Table 2 presents the complete empirical results for the Global Friendship Paradox evaluation across all six centrality measures and all five real-world network datasets (*Karate Club*, *Florentine*

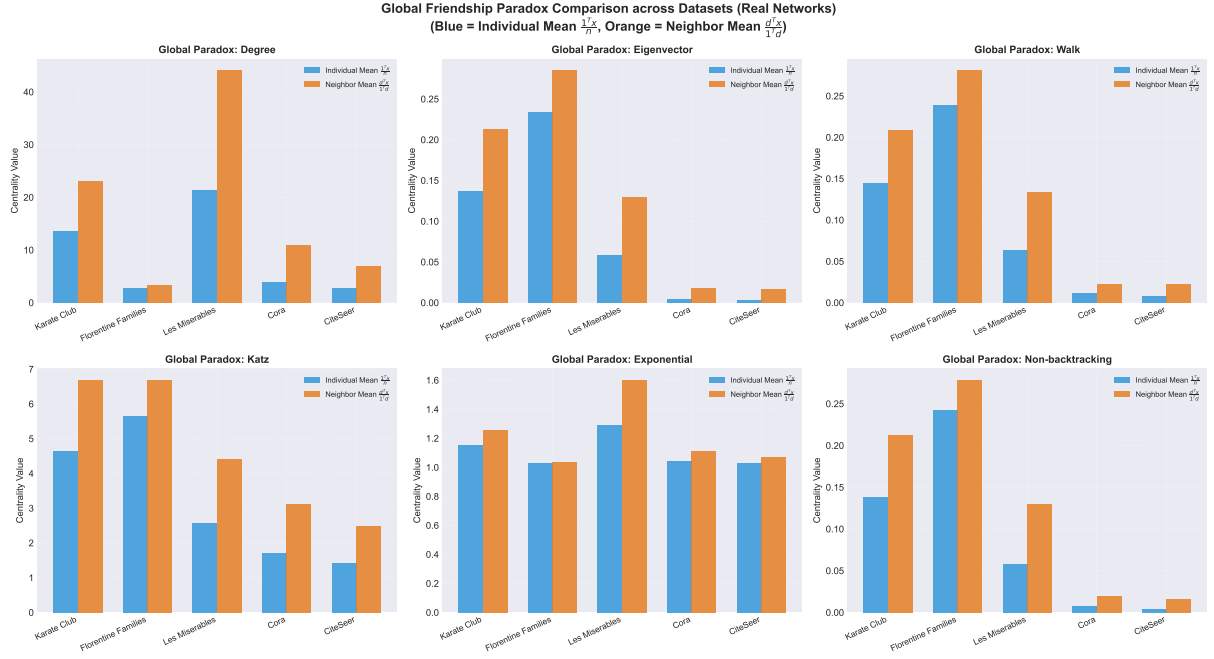


Figure 1: Global Friendship Paradox Comparison across Datasets for Real Networks.

Table 2: Global Friendship Paradox evaluation across all networks and centralities.

Graph	Centrality	Real $\mathbb{E}[x]$	Real $\mathbb{E}[x_{\text{neigh}}]$	Real Δ_{GFP}	Random $\mathbb{E}[x]$	Random $\mathbb{E}[x_{\text{neigh}}]$	Random Δ_{GFP}
Karate Club	Degree	13.5882	23.0705	9.4823	4.7647	5.3580	0.5933
Karate Club	Eigenvector	0.1374	0.2125	0.0751	0.1543	0.1787	0.0244
Karate Club	Walk	0.1449	0.2084	0.0635	0.1591	0.1806	0.0214
Karate Club	Katz	4.6542	6.6852	2.0310	5.6663	6.3580	0.6917
Karate Club	Exponential	1.1531	1.2556	0.1025	1.0489	1.0551	0.0061
Karate Club	Non-backtracking	0.1378	0.2121	0.0742	0.1596	0.1800	0.0204
Florentine Families	Degree	2.6667	3.3500	0.6833	2.5333	3.2632	0.7298
Florentine Families	Eigenvector	0.2335	0.2851	0.0516	0.2280	0.2887	0.0607
Florentine Families	Walk	0.2391	0.2810	0.0419	0.2338	0.2888	0.0550
Florentine Families	Katz	5.6537	6.6851	1.0314	5.4543	6.6341	1.1798
Florentine Families	Exponential	1.0271	1.0340	0.0069	1.0258	1.0331	0.0074
Florentine Families	Non-backtracking	0.2421	0.2783	0.0363	0.2409	0.2860	0.0451
Les Miserables	Degree	21.2987	44.0886	22.7899	6.4156	7.0972	0.6816
Les Miserables	Eigenvector	0.0584	0.1295	0.0711	0.1053	0.1181	0.0129
Les Miserables	Walk	0.0637	0.1333	0.0695	0.1080	0.1190	0.0111
Les Miserables	Katz	2.5664	4.4042	1.8378	5.8980	6.4659	0.5678
Les Miserables	Exponential	1.2885	1.6007	0.3122	1.0665	1.0735	0.0071
Les Miserables	Non-backtracking	0.0584	0.1294	0.0710	0.1076	0.1187	0.0111
Cora	Degree	3.8981	10.9092	7.0112	3.8530	4.8491	0.9961
Cora	Eigenvector	0.0048	0.0177	0.0129	0.0153	0.0202	0.0049
Cora	Walk	0.0116	0.0228	0.0112	0.0169	0.0212	0.0043
Cora	Katz	1.7183	3.1200	1.4016	4.8549	5.9866	1.1317
Cora	Exponential	1.0412	1.1135	0.0723	1.0395	1.0497	0.0102
Cora	Non-backtracking	0.0075	0.0194	0.0118	0.0166	0.0209	0.0042
CiteSeer	Degree	2.7364	6.9134	4.1770	2.6619	3.6188	0.9569
CiteSeer	Eigenvector	0.0034	0.0170	0.0136	0.0093	0.0140	0.0047
CiteSeer	Walk	0.0077	0.0220	0.0144	0.0145	0.0197	0.0052
CiteSeer	Katz	1.4204	2.4843	1.0639	3.8667	5.0678	1.2012
CiteSeer	Exponential	1.0283	1.0720	0.0436	1.0271	1.0368	0.0097
CiteSeer	Non-backtracking	0.0036	0.0164	0.0128	0.0140	0.0189	0.0049

Families, *Les Misérables*, *Cora*, and *CiteSeer*) compared against their synthetic Erdős–Rényi null models.

As shown in Table 2 and illustrated in Figure 1, the global GFP inequality $\mathbb{E}[x_{\text{neigh}}] \geq \mathbb{E}[x]$ holds across all evaluated real-world networks and centralities. The magnitude of the global gap Δ_{GFP} , however, depends heavily on the underlying network structure. In highly heterogeneous networks, Δ_{GFP} is substantially larger in real topologies than in Erdős–Rényi graphs. For instance, in *Karate Club* under Degree centrality, $\Delta_{\text{GFP}} = 9.4823$ in the real graph compared to only 0.5933 in the random model. Similarly, in *Les Misérables* under Degree centrality, the gap reaches 22.7899 in the empirical graph compared to 0.6816 in the random network.

Conversely, in networks with lower degree variance or smaller scales, such as *Florentine Families* across all measures or *CiteSeer* under Katz centrality, the random model exhibits a slightly higher gap than its empirical counterpart.

Amplification via Heterogeneity : Degree variance and heavy-tailed structural heterogeneity in empirical networks drastically expand the global gap Δ_{GFP} relative to random null models, turning the Global Friendship Paradox into a significantly stronger effect in real-world graphs.

6 The Local Friendship Gap Δ_i

6.1 Definition

Local Friendship Gap Δ_i . For a given node i , the local friendship gap Δ_i measures the relative perception bias experienced locally by node i . In matrix form across the entire network:

$$\Delta = D^{-1}A\mathbf{x} - \mathbf{x} \quad (19)$$

Equivalently, for a specific node i , the local gap is written element-wise as:

$$\Delta_i = (D^{-1}A\mathbf{x})_i - x_i = \frac{1}{d_i} \sum_{j=1}^n A_{i,j}x_j - x_i \quad (20)$$

where $A_{i,j}$ represents the entries of the adjacency matrix A , $D = \text{diag}(d_1, \dots, d_n)$ is the diagonal degree matrix with $d_i = \sum_j A_{i,j}$, and $\mathbf{x} = (x_1, \dots, x_n)^\top$ is the vector of node importances.

Figure 2 provides a structural illustration of this local gap centered on Node 1 of the *Karate Club* network alongside its immediate 1-hop neighborhood. To analyze the local perception bias for Node 1 across different topological scales, its specific centrality values x_1 are summarized in Table 3.

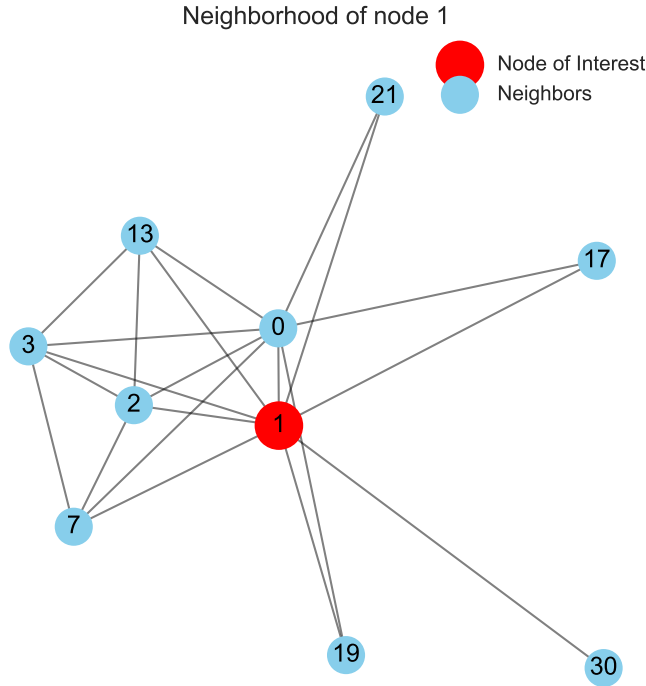


Figure 2: Structural representation of the local neighborhood surrounding Node 1 in the *Karate Club* network (Red: Node of Interest, Light Blue: Immediate 1-hop Neighbors).

6.2 Interpretation of Δ_i

Table 3: Centrality metrics for Node 1 across topological scales.

Centrality Measure	Value (x_1)
Degree	9
Eigenvector	0.2660
Random Walk ($l = 2$)	600.0
Katz	0.1972
Exponential	11.3713
Non-backtracking	0.3023

Sign Interpretation of Δ_i

Intuition:

- $\Delta_i > 0$ (**Local Paradox**): Node i suffers from local perception bias—on average, its immediate neighbors possess a higher centrality score than node i itself.
- $\Delta_i < 0$ (**Dominant Node**): Node i is locally dominant, its neighbors are less central on average than node i itself.

6.3 Empirical Distribution of Δ_i

The continuous density distributions of Δ_i across networks highlight distinct behaviors between empirical topologies and random networks.

Figure 3 illustrates the continuous density distribution of Δ_i for the *Florentine Families* network across six centralities. Because *Florentine Families* is a small, relatively homogeneous network ($\text{Var}(d) = 1.82$ in real vs 1.85 in random), the distributions for the real network (red) and the Erdős-Rényi graph (blue) closely mirror each other. In both cases, the peak lies to the right of $\Delta_i = 0$, confirming that a majority of nodes experience $\Delta_i \geq 0$ (80.00% in real vs 64.29% in random for degree centrality).

For the *Karate Club* network, shown in Figure 4, the higher structural heterogeneity ($\text{Var}(d) = 14.60$ in real vs 2.83 in random) causes a significant rightward shift in the real density curve compared to the random model.

Figure 5 depicts the distribution for *Les Miserables*. Here, strong clustering and core-periphery dynamics ($\text{Var}(d) = 36.01$ in real vs 4.37 in random) amplify the local perception gap across degree, eigenvector, and non-backtracking centralities.

In larger citation networks, as illustrated in Figure 6 (*Cora*) and Figure 7 (*CiteSeer*), the proportion of paradoxical nodes under spectral and non-backtracking centralities reaches particularly high levels.

To visually compare the prevalence of the local paradox across datasets and centrality metrics, Figure 8 presents a comprehensive heatmap of the percentage of paradoxical nodes ($\%\Delta_i \geq 0$) comparing empirical topologies to their random counterparts.

Table 4 provides the complete empirical summary of the expected local gap $\mathbb{E}[\Delta_i]$ and the percentage of paradoxical nodes ($\%\Delta_i \geq 0$) across all five real-world networks and six centralities, along with degree variance statistics.

In larger real-world networks such as *Cora* and *CiteSeer*, the proportion of paradoxical nodes under eigenvector centrality reaches high levels (93.91% for *Cora* and 81.49% for *CiteSeer*, as detailed in Table 4 and Figure 8). Non-backtracking centrality yields similar results (92.17% for *Cora* and 81.31% for *CiteSeer*).

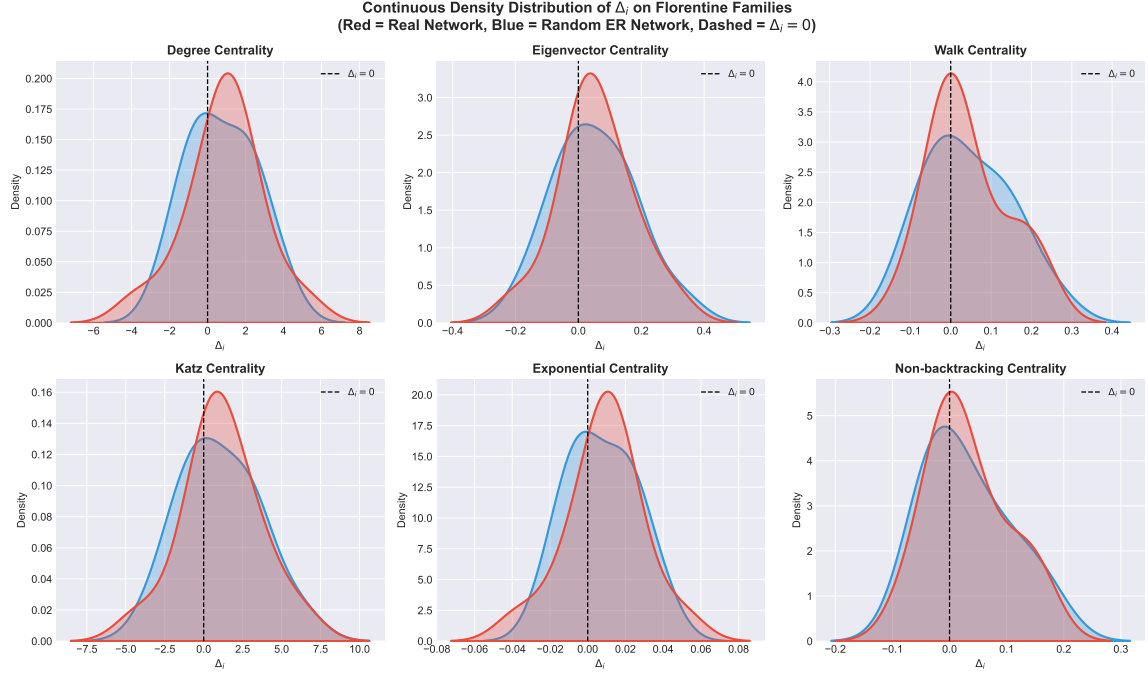


Figure 3: Continuous density distribution of local gap Δ_i on the *Florentine Families* network across six centrality measures (Red: Real Network, Blue: Random ER Network, Dashed line: $\Delta_i = 0$).

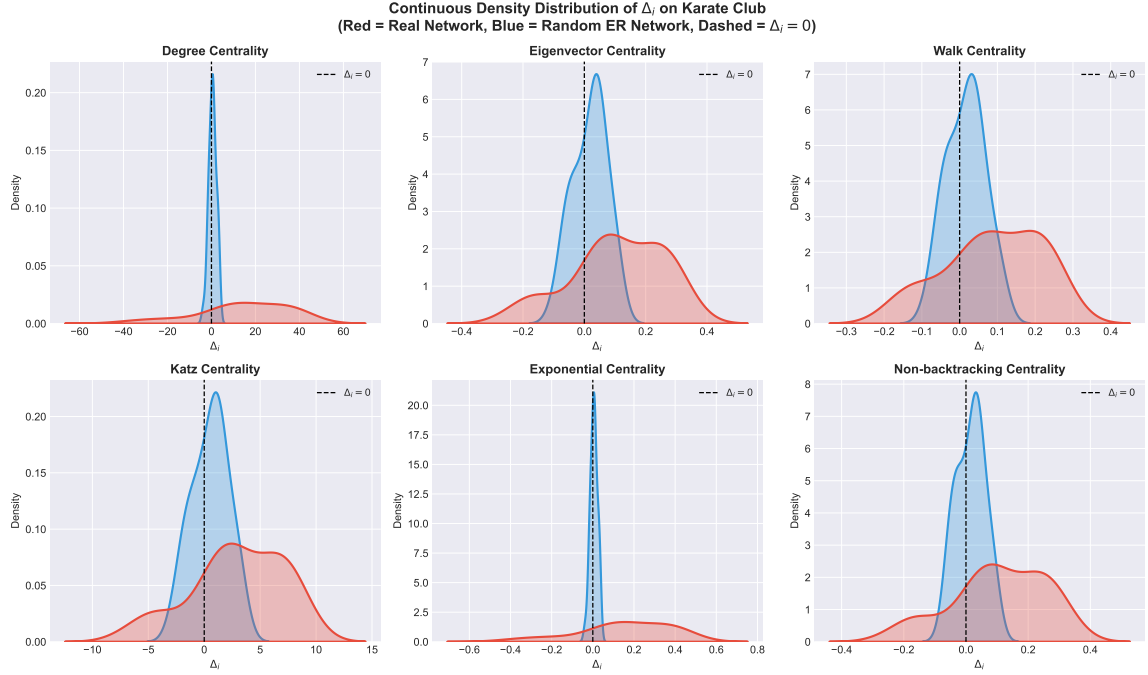


Figure 4: Continuous density distribution of local gap Δ_i on the *Karate Club* network across six centrality measures (Red: Real Network, Blue: Random ER Network, Dashed line: $\Delta_i = 0$).

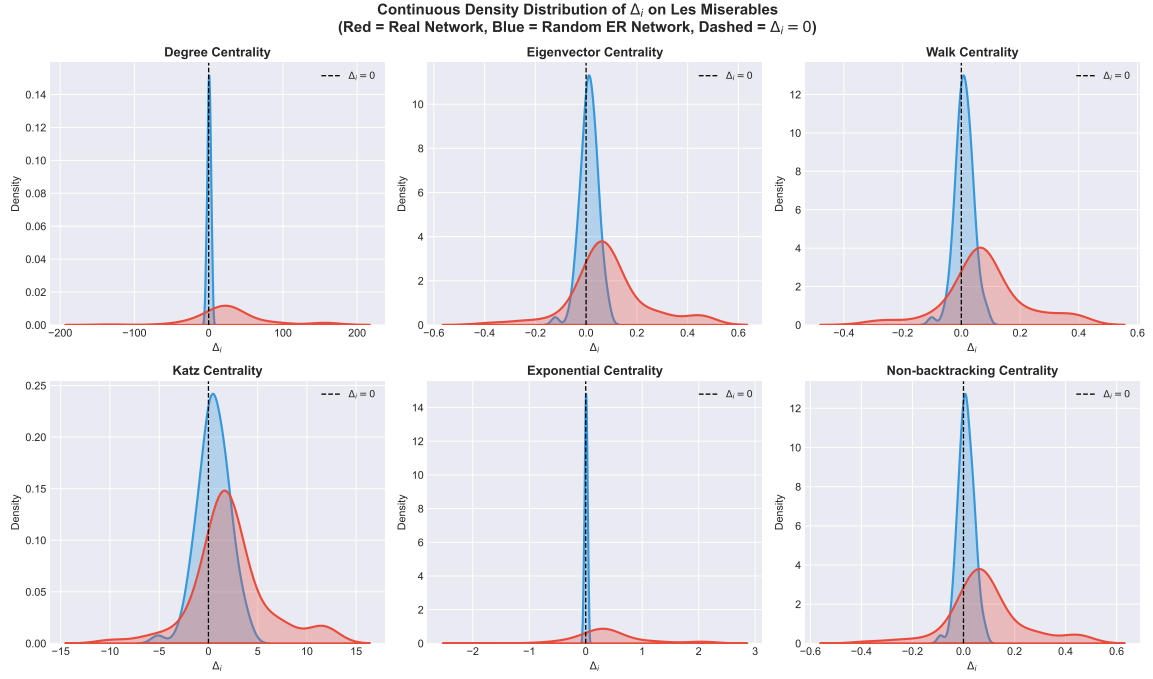


Figure 5: Continuous density distribution of local gap Δ_i on the *Les Miserables* network across six centrality measures (Red: Real Network, Blue: Random ER Network, Dashed line: $\Delta_i = 0$).

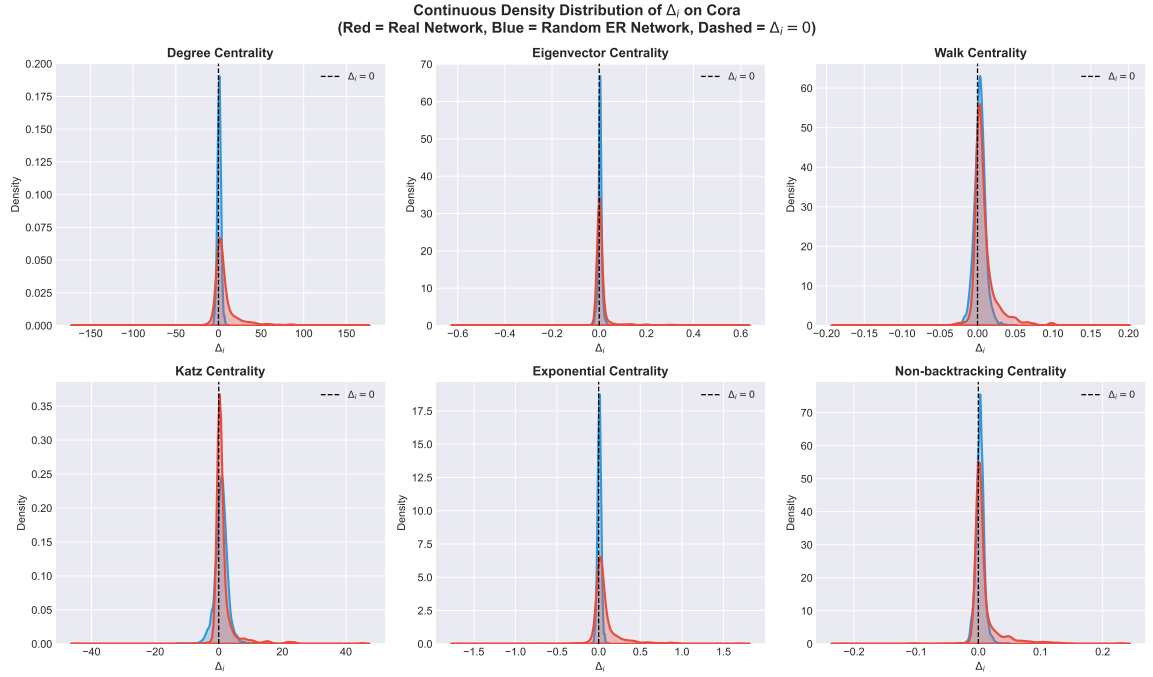


Figure 6: Continuous density distribution of local gap Δ_i on the *Cora* network across six centrality measures (Red: Real Network, Blue: Random ER Network, Dashed line: $\Delta_i = 0$).

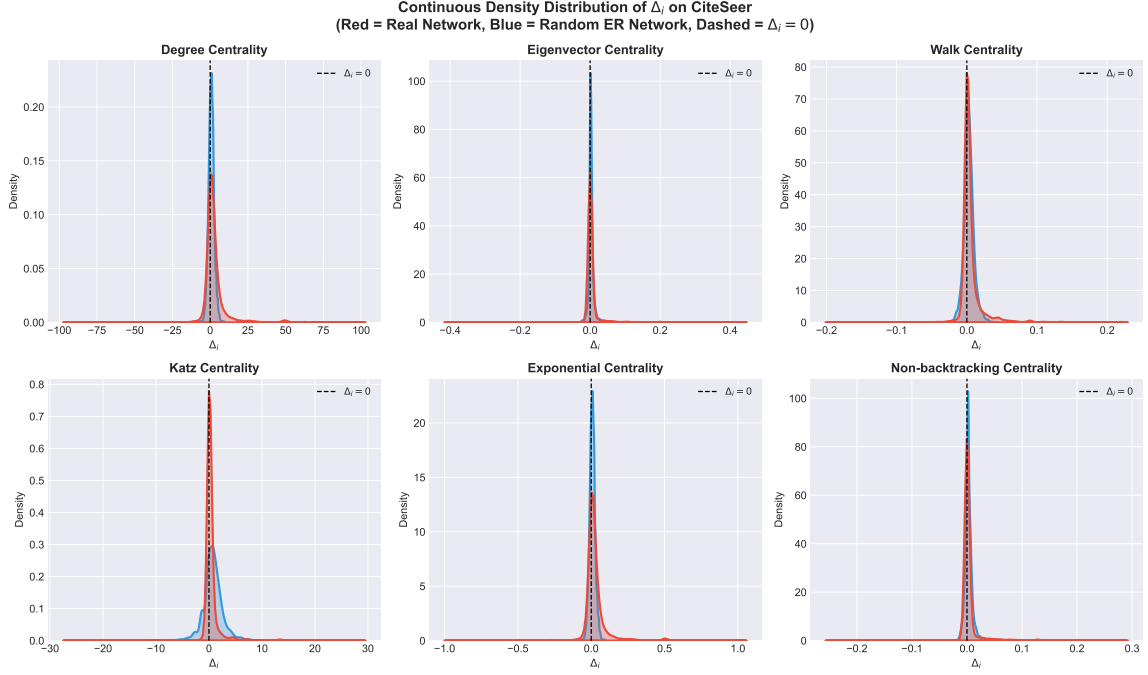


Figure 7: Continuous density distribution of local gap Δ_i on the *CiteSeer* network across six centrality measures (Red: Real Network, Blue: Random ER Network, Dashed line: $\Delta_i = 0$).

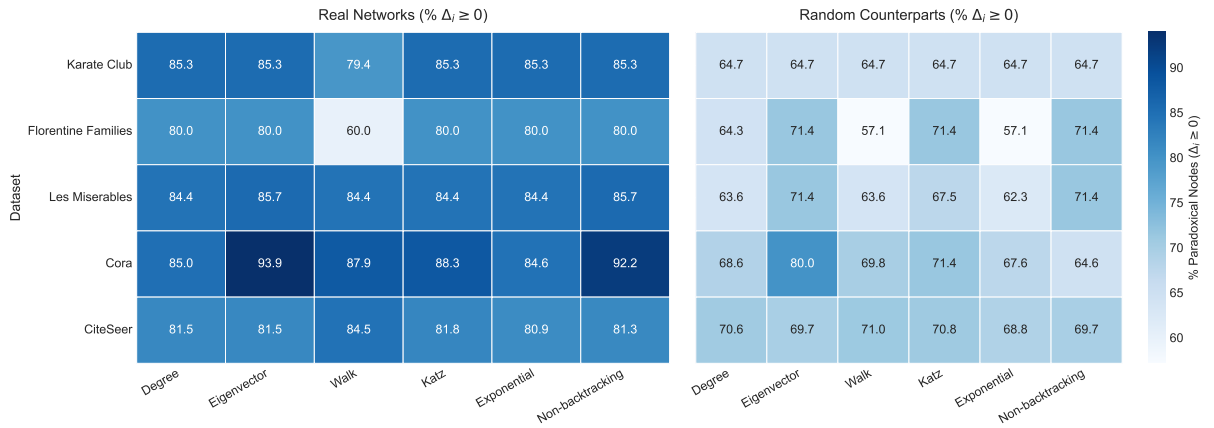


Figure 8: Heatmap of the Local Friendship Paradox prevalence across empirical datasets and centralities (Percentage of nodes with $\Delta_i \geq 0$ in Real vs. Random counterparts).

Table 4: Summary of local gap metrics $\mathbb{E}[\Delta_i]$, $\% \Delta_i \geq 0$, and degree variances across real and random networks.

Graph	Centrality	Real $\% \Delta_i \geq 0$	Random $\% \Delta_i \geq 0$	Real $\mathbb{E}[\Delta_i]$	Random $\mathbb{E}[\Delta_i]$	Real $\text{Var}(d)$	Random $\text{Var}(d)$
Karate Club	Degree	85.29%	64.71%	14.7383	0.5403	14.60	2.83
Karate Club	Eigenvector	85.29%	64.71%	0.1065	0.0186	14.60	2.83
Karate Club	Walk	79.41%	64.71%	0.0881	0.0168	14.60	2.83
Karate Club	Katz	85.29%	64.71%	2.9337	0.5462	14.60	2.83
Karate Club	Exponential	85.29%	64.71%	0.1581	0.0055	14.60	2.83
Karate Club	Non-backtracking	85.29%	64.71%	0.1050	0.0155	14.60	2.83
Florentine Families	Degree	80.00%	64.29%	0.8778	0.7476	1.82	1.85
Florentine Families	Eigenvector	80.00%	71.43%	0.0549	0.0531	1.82	1.85
Florentine Families	Walk	60.00%	57.14%	0.0425	0.0441	1.82	1.85
Florentine Families	Katz	80.00%	71.43%	1.1433	1.0568	1.82	1.85
Florentine Families	Exponential	80.00%	57.14%	0.0089	0.0076	1.82	1.85
Florentine Families	Non-backtracking	80.00%	71.43%	0.0318	0.0319	1.82	1.85
Les Miserables	Degree	84.42%	63.64%	29.7558	0.7209	36.01	4.37
Les Miserables	Eigenvector	85.71%	71.43%	0.0871	0.0113	36.01	4.37
Les Miserables	Walk	84.42%	63.64%	0.0817	0.0098	36.01	4.37
Les Miserables	Katz	84.42%	67.53%	2.2657	0.5159	36.01	4.37
Les Miserables	Exponential	84.42%	62.34%	0.3977	0.0074	36.01	4.37
Les Miserables	Non-backtracking	85.71%	71.43%	0.0868	0.0098	36.01	4.37
Cora	Degree	84.97%	68.65%	7.7337	0.9040	27.33	3.84
Cora	Eigenvector	93.91%	79.97%	0.0126	0.0035	27.33	3.84
Cora	Walk	87.92%	69.78%	0.0091	0.0032	27.33	3.84
Cora	Katz	88.26%	71.41%	1.3983	0.8544	27.33	3.84
Cora	Exponential	84.56%	67.55%	0.0792	0.0092	27.33	3.84
Cora	Non-backtracking	92.17%	64.63%	0.0098	0.0029	27.33	3.84
CiteSeer	Degree	81.52%	70.63%	2.8014	0.7732	11.43	2.55
CiteSeer	Eigenvector	81.49%	69.69%	0.0053	0.0028	11.43	2.55
CiteSeer	Walk	84.51%	70.98%	0.0055	0.0031	11.43	2.55
CiteSeer	Katz	81.82%	70.82%	0.4999	0.7743	11.43	2.55
CiteSeer	Exponential	80.94%	68.79%	0.0287	0.0078	11.43	2.55
CiteSeer	Non-backtracking	81.31%	69.69%	0.0042	0.0025	11.43	2.55

6.4 Empirical Gaussian Approximation in Erdős–Rényi Baselines

The primary objective of this subsection is to evaluate whether the local perception gap Δ_i in random network baselines can be reasonably approximated by a continuous Gaussian distribution $\mathcal{N}(\mu, \sigma^2)$. Under the classic Erdős–Rényi (*ER*) baseline model $G(n, p)$, edge formation is independent and homogeneous, resulting in binomial degree distributions bounded near the mean. Although local node-level observations of Δ_i involve dependent neighborhood sums, the overall distribution of Δ_i in dense or moderately connected *ER* graphs empirically displays a symmetric, bell-shaped profile. We evaluate this Gaussian model as a baseline reference for comparing real-world structural deviations.

Methodology and Statistical Formulation. For each baseline network topology, we isolate the node-level observations of the local perception gap under Degree Centrality using the adjacency matrix representation.

Data Preprocessing Note on Graph Sizes (N) : In evaluating Δ_i , isolated nodes ($d_i = 0$) are excluded to prevent division by zero in average neighbor computations. Consequently, for networks containing isolated or component-disconnected vertices in the random baseline realization (such as *Cora* or *CiteSeer*), the effective sample size N of analyzed nodes in the *ER* model may be slightly lower than the total vertex count of the original graph.

To model Δ_i as a continuous benchmark, we perform Maximum Likelihood Estimation (MLE) to estimate the mean $\hat{\mu}$ and standard deviation $\hat{\sigma}$ of the fitted normal density across the N active nodes.

Parametric Maximum Likelihood Estimation (MLE)

$$\hat{\mu} = \frac{1}{N} \sum_{i=1}^N \Delta_i, \quad \hat{\sigma} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\Delta_i - \hat{\mu})^2} \quad (21)$$

Intuition: $\hat{\mu}$ measures the network-wide mean perception gap, while $\hat{\sigma}$ reflects the dispersion of individual perception gaps around this baseline average.

The resulting parametric Gaussian probability density function $f(\Delta_i; \hat{\mu}, \hat{\sigma})$ is defined as:

Gaussian Baseline Probability Density Function $f(\Delta_i; \hat{\mu}, \hat{\sigma})$

$$f(\Delta_i; \hat{\mu}, \hat{\sigma}) = \frac{1}{\hat{\sigma}\sqrt{2\pi}} \exp\left(-\frac{1}{2} \left(\frac{\Delta_i - \hat{\mu}}{\hat{\sigma}}\right)^2\right) \quad (22)$$

Intuition: This density serves as an analytical reference distribution to quantify whether perception gaps behave similarly to symmetric, unclustered baseline fluctuations.

To quantify structural asymmetry and deviation from Gaussian symmetry, we compute the sample skewness γ_1 .

Sample Skewness γ_1

$$\gamma_1 = \frac{\frac{1}{N} \sum_{i=1}^N (\Delta_i - \hat{\mu})^3}{\hat{\sigma}^3} \quad (23)$$

Intuition: Skewness measures distributional asymmetry. Values near 0 indicate a balanced bell shape, whereas a strong positive skew ($\gamma_1 \gg 0$) highlights heavy right-tail behavior associated with high-degree hubs inflating local perception gaps.

Finally, to quantify deviation from the fitted Gaussian baseline, we compute the empirical cumulative distribution function (ECDF) $F_N(x)$ and evaluate the maximum divergence.

Empirical Cumulative Distribution Function (ECDF) $F_N(x)$

$$F_N(x) = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\Delta_i \leq x) \quad (24)$$

Intuition: The ECDF aggregates empirical observations, tracking the cumulative fraction of nodes with a perception gap less than or equal to x .

The Kolmogorov–Smirnov (KS) distance statistic D quantifies the supremum distance between $F_N(x)$ and the fitted Gaussian cumulative distribution function $\Phi(x; \hat{\mu}, \hat{\sigma})$.

Descriptive Kolmogorov–Smirnov Distance Statistic (D)

$$D = \sup_{x \in \mathbb{R}} \left| F_N(x) - \Phi\left(\frac{x - \hat{\mu}}{\hat{\sigma}}\right) \right| \quad (25)$$

Intuition: The D statistic serves as a descriptive metric measuring the maximum vertical discrepancy between the empirical cumulative data distribution (F_N) and the Gaussian

baseline model ($\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-t^2/2} dt$). Smaller values of D indicate closer alignment with a normal reference profile.

All computed statistical metrics across random baselines and empirical networks are summarized in Table 5.

Empirical Results and Discussion. The empirical distributions for Erdős–Rényi random baselines alongside their corresponding fitted Gaussian references are shown for small-scale networks in Figure 9 (*Karate Club*, $\hat{\mu} = 0.5403, \hat{\sigma} = 1.6768$), Figure 10 (*Florentine Families*, $\hat{\mu} = 0.7476, \hat{\sigma} = 1.7551$), and Figure 11 (*Les Misérables*, $\hat{\mu} = 0.7209, \hat{\sigma} = 2.3764$).

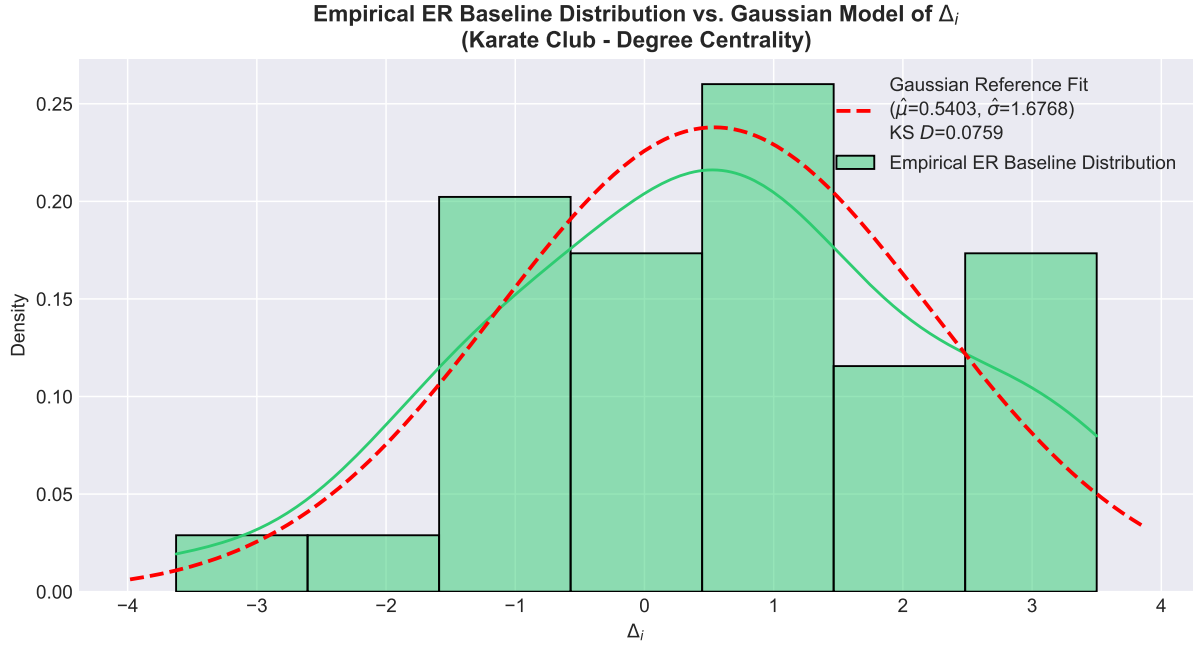


Figure 9: Empirical ER baseline distribution vs. fitted parametric normal model for Δ_i on *Karate Club* (Degree Centrality; Green: Empirical ER Baseline, Red Dashed: Gaussian Reference $\hat{\mu} = 0.5403, \hat{\sigma} = 1.6768$, KS $D = 0.0759$).

In these small random graphs ($N \leq 77$), the empirical distributions exhibit minimal skewness (between -0.1494 and 0.1221) and low KS distances ($D \leq 0.1649$), confirming that the empirical distribution of Δ_i in homogeneous *ER* baselines remains close to a Gaussian density profile.

For larger citation baselines, shown in Figure 12 (*Cora*, $\hat{\mu} = 0.9040, \hat{\sigma} = 2.2164$) and Figure 13 (*CiteSeer*, $\hat{\mu} = 0.7732, \hat{\sigma} = 1.8549$), the empirical profile maintains near-zero skewness (0.1139 and 0.1847 , respectively) and tight visual symmetry around the mean. The low distance statistics ($D = 0.0473$ and $D = 0.0786$) demonstrate that the Gaussian distribution remains a strong analytical approximation for random baseline perception gaps.

6.5 Non-Gaussian Dynamics and Structural Deviations in Real-World Networks

Objective. In contrast to the Erdős–Rényi baseline where Δ_i exhibits symmetric and bounded fluctuations around small positive means, real-world networks feature complex organization, including high degree heterogeneity, community structure, and prominent hubs. The objective of this subsection is to quantify how empirical distributions of Δ_i in real networks systematically deviate from Gaussian reference models, illustrating the impact of real structural heterogeneity.

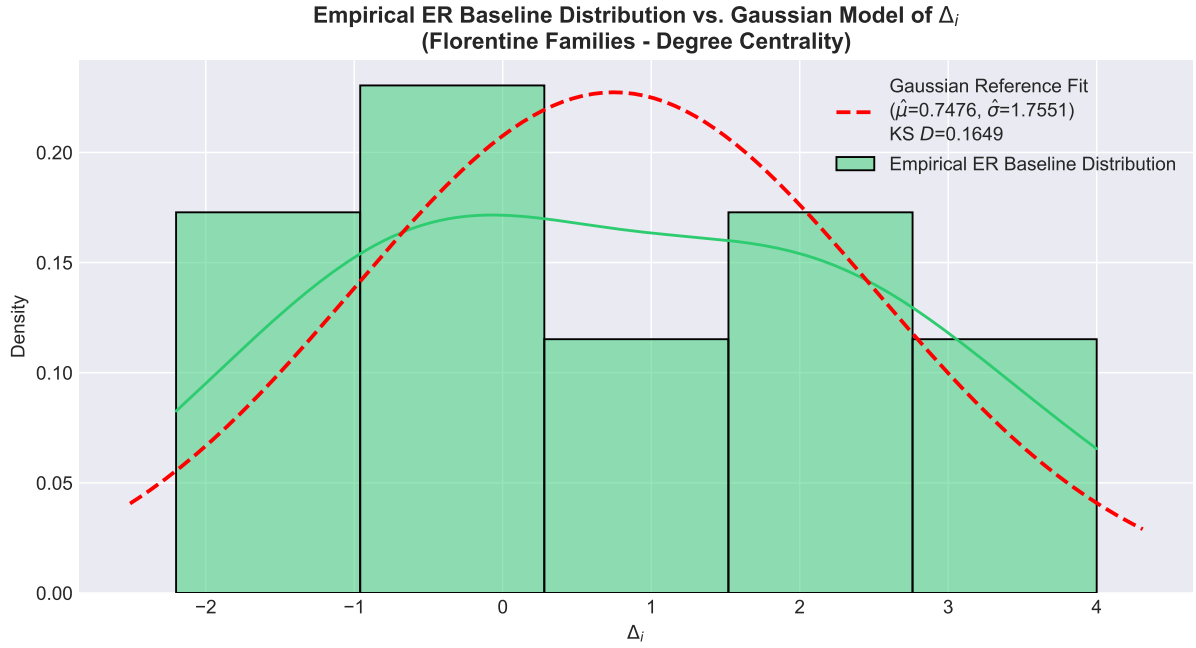


Figure 10: Empirical ER baseline distribution vs. fitted parametric normal model for Δ_i on *Florentine Families* (Degree Centrality; Green: Empirical ER Baseline, Red Dashed: Gaussian Reference $\hat{\mu} = 0.7476, \hat{\sigma} = 1.7551$, KS $D = 0.1649$).

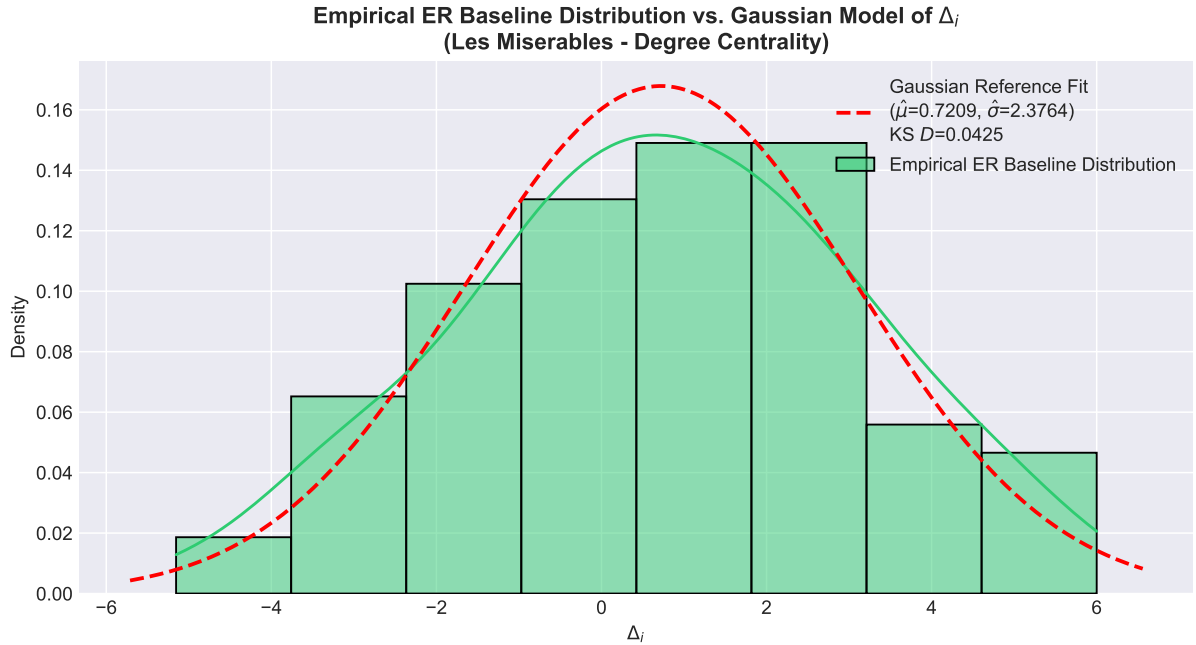


Figure 11: Empirical ER baseline distribution vs. fitted parametric normal model for Δ_i on *Les Miserables* (Degree Centrality; Green: Empirical ER Baseline, Red Dashed: Gaussian Reference $\hat{\mu} = 0.7209, \hat{\sigma} = 2.3764$, KS $D = 0.0425$).

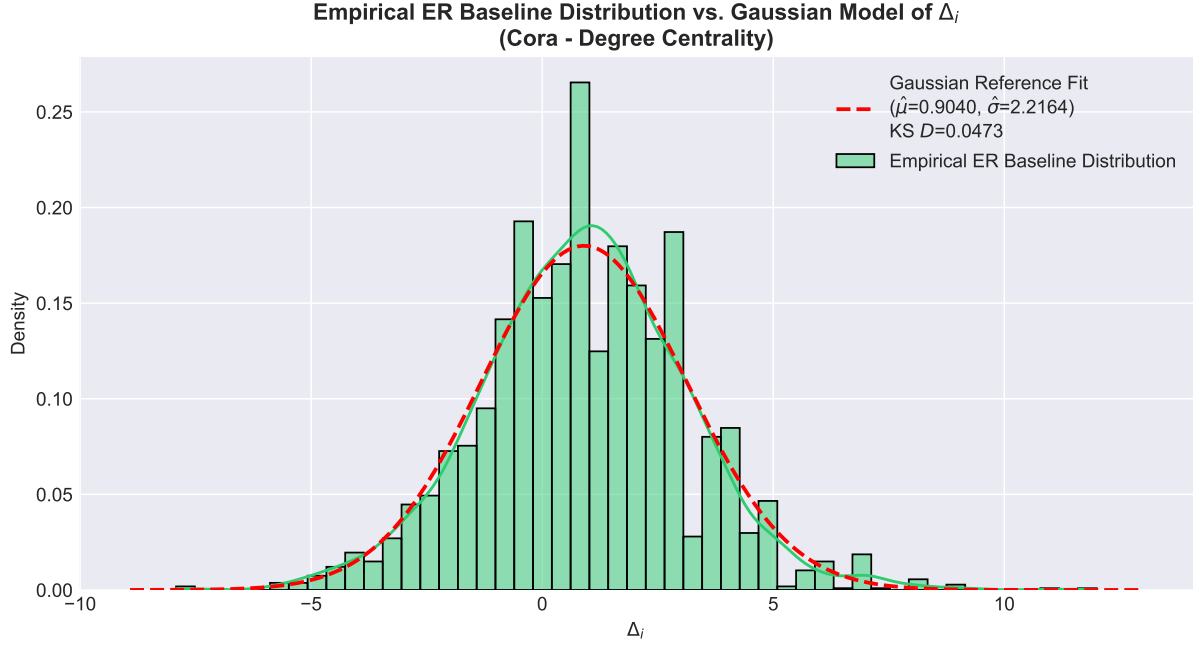


Figure 12: Empirical ER baseline distribution vs. fitted parametric normal model for Δ_i on *Cora* (Degree Centrality; Green: Empirical ER Baseline, Red Dashed: Gaussian Reference $\hat{\mu} = 0.9040$, $\hat{\sigma} = 2.2164$, KS $D = 0.0473$).

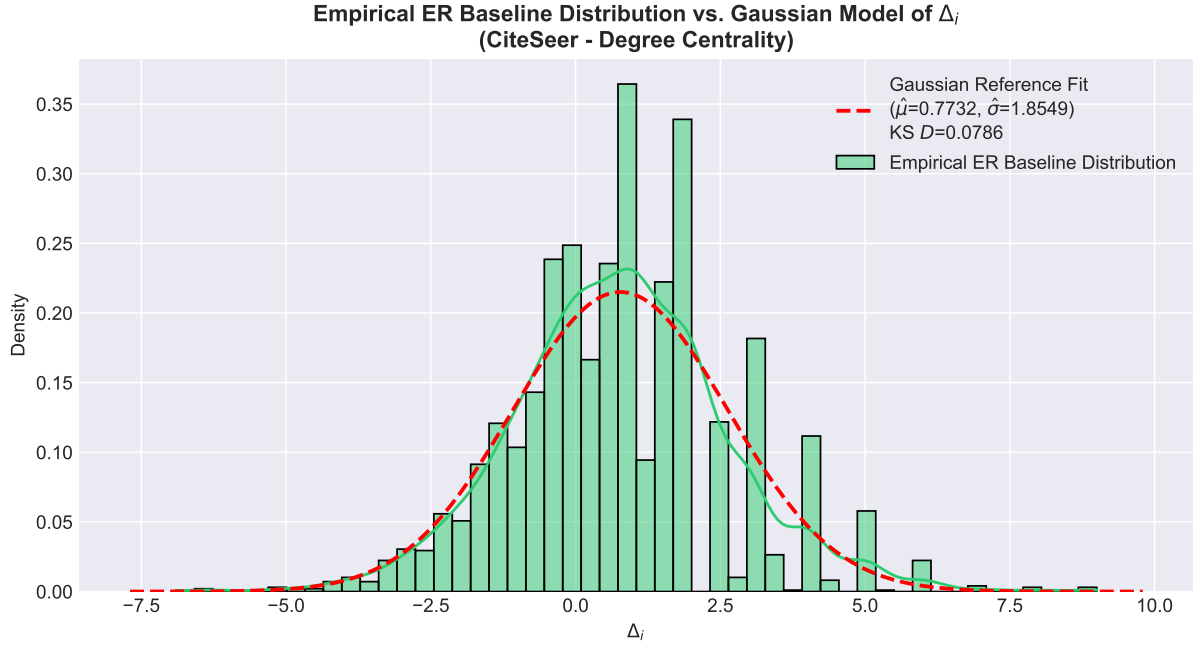


Figure 13: Empirical ER baseline distribution vs. fitted parametric normal model for Δ_i on *CiteSeer* (Degree Centrality; Green: Empirical ER Baseline, Red Dashed: Gaussian Reference $\hat{\mu} = 0.7732$, $\hat{\sigma} = 1.8549$, KS $D = 0.0786$).

Methodology and Procedure. For each empirical dataset, node-level observations of Δ_i under Degree Centrality are fitted to a Gaussian reference distribution $\mathcal{N}(\hat{\mu}_{\text{real}}, \hat{\sigma}_{\text{real}}^2)$ via MLE:

$$f(\Delta_i; \hat{\mu}_{\text{real}}, \hat{\sigma}_{\text{real}}) = \frac{1}{\hat{\sigma}_{\text{real}}\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{\Delta_i - \hat{\mu}_{\text{real}}}{\hat{\sigma}_{\text{real}}}\right)^2\right) \quad (26)$$

Structural deviations from this reference model are described using the Kolmogorov–Smirnov distance (D_{real}), sample skewness, and overall paradox prevalence ($\%\Delta_i \geq 0$), as summarized in Table 5.

Empirical Results and Discussion. The empirical distributions for small real-world networks alongside their Gaussian reference fits are illustrated in Figure 14 (*Karate Club*), Figure 15 (*Florentine Families*), and Figure 16 (*Les Miserables*).

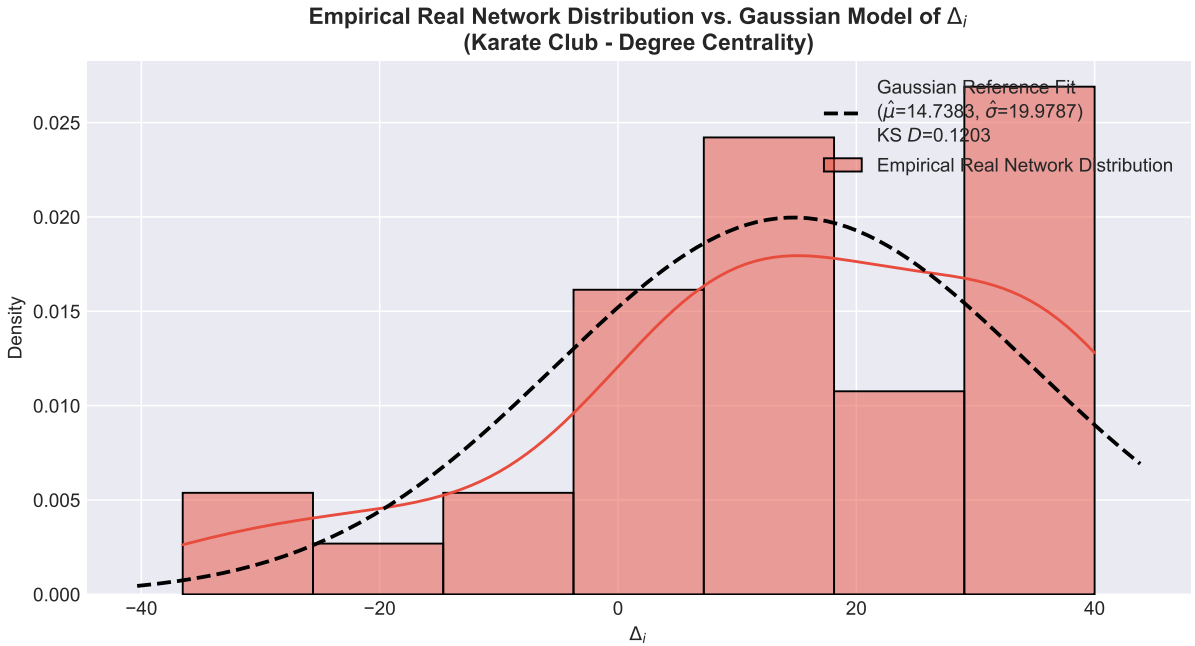


Figure 14: Empirical real-world distribution vs. fitted Gaussian reference for Δ_i on *Karate Club* (Degree Centrality; Red: Empirical Real, Black Dashed: Gaussian Reference $\hat{\mu} = 14.7383, \hat{\sigma} = 19.9787$, KS $D = 0.1203$).

As shown in Table 5, real networks systematically yield higher mean perception gaps ($\hat{\mu}$) and substantially larger standard deviations ($\hat{\sigma}$) than their ER baseline counterparts. For instance, in *Les Miserables* (Figure 16), $\hat{\mu}$ increases from 0.7209 in the ER baseline to 29.7558 in the real network, while standard deviation increases from $\hat{\sigma} = 2.3764$ to 47.4002, accompanied by a notable increase in distance D to 0.1863.

This non-Gaussian behavior is particularly evident in large citation networks, shown in Figure 17 (*Cora*) and Figure 18 (*CiteSeer*).

6.6 Comparative Synthesis: Structural Drivers of the Perception Gap

The quantitative comparison in Table 5 demonstrates clear differences in how local perception gaps behave across network classes:

1. **Systematic Amplification of Paradox Prevalence:** In homogeneous Erdős–Rényi baselines, the proportion of nodes experiencing a positive perception gap ($\%\Delta_i \geq 0$) remains

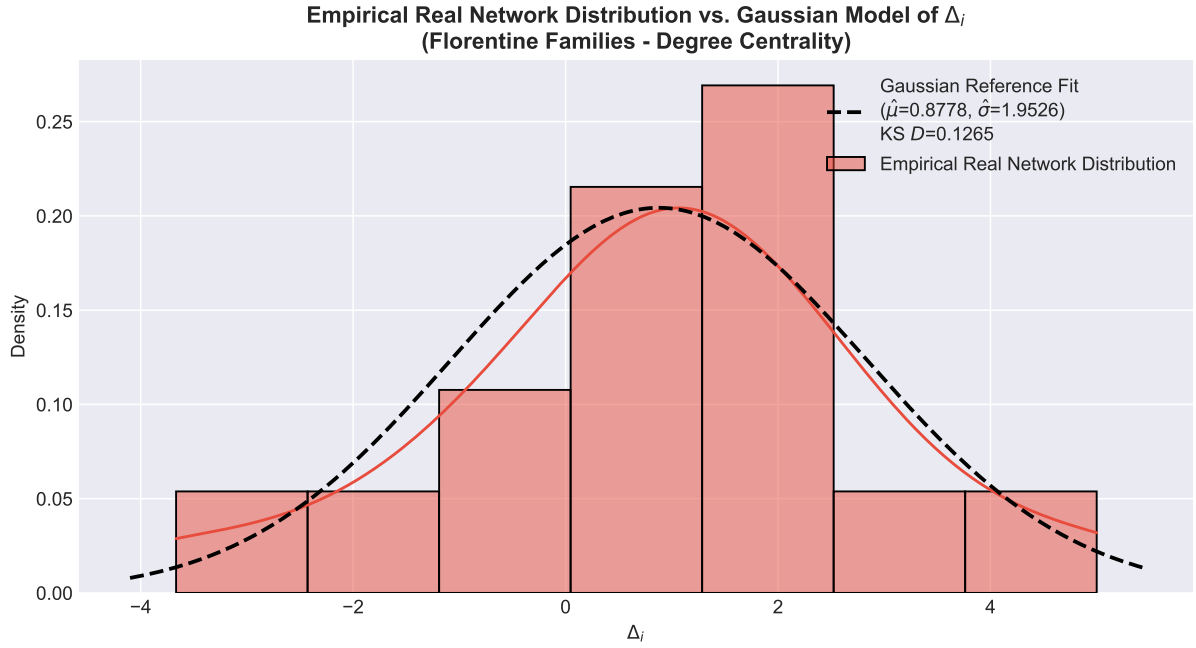


Figure 15: Empirical real-world distribution vs. fitted Gaussian reference for Δ_i on *Florentine Families* (Degree Centrality; Red: Empirical Real, Black Dashed: Gaussian Reference $\hat{\mu} = 0.8778, \hat{\sigma} = 1.9526$, KS $D = 0.1265$).

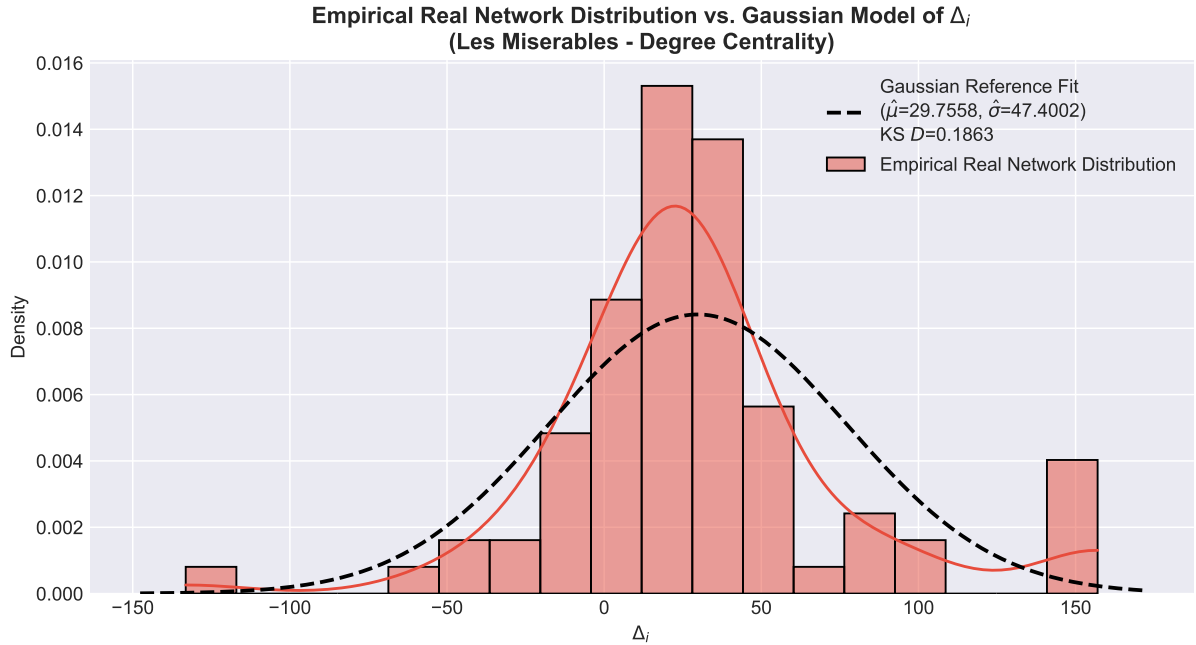


Figure 16: Empirical real-world distribution vs. fitted Gaussian reference for Δ_i on *Les Miserables* (Degree Centrality; Red: Empirical Real, Black Dashed: Gaussian Reference $\hat{\mu} = 29.7558, \hat{\sigma} = 47.4002$, KS $D = 0.1863$).

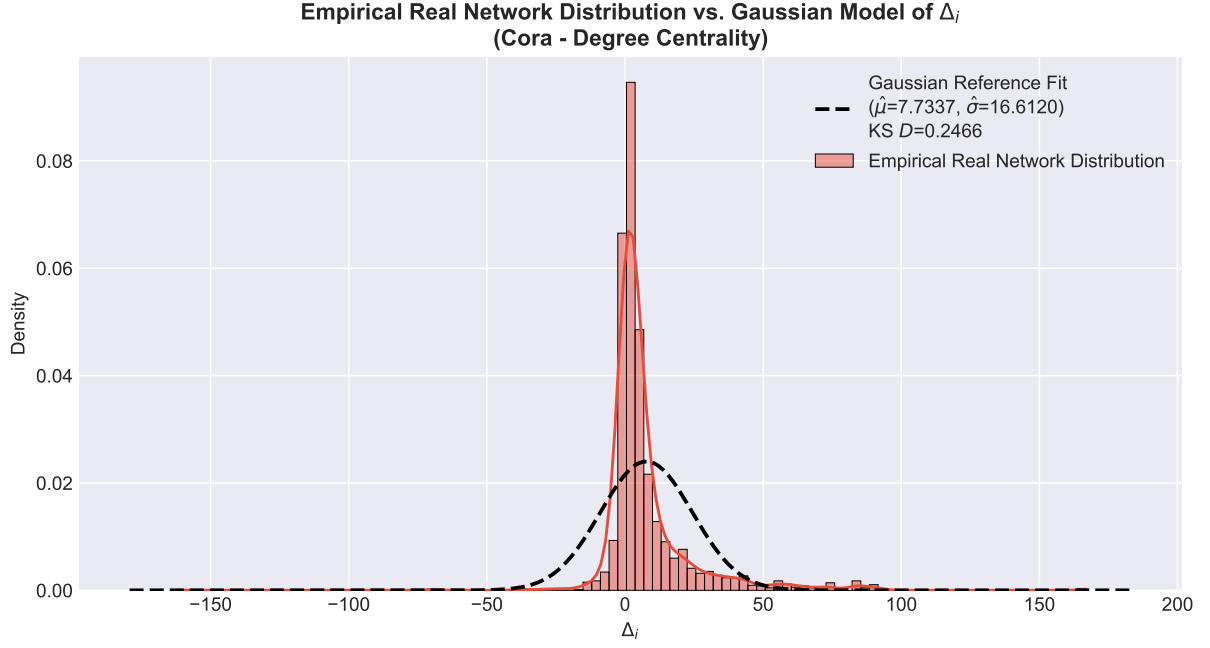


Figure 17: Empirical real-world distribution vs. fitted Gaussian reference for Δ_i on *Cora* (Degree Centrality; Red: Empirical Real, Black Dashed: Gaussian Reference $\hat{\mu} = 7.7337, \hat{\sigma} = 16.6120$, KS $D = 0.2466$).

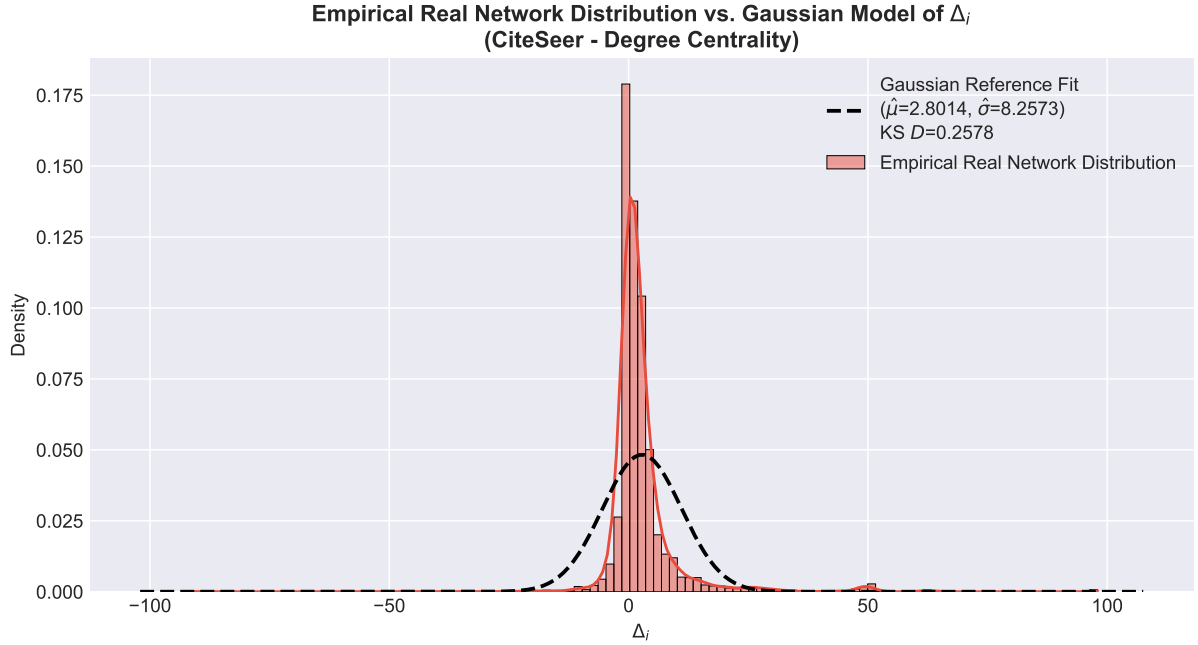


Figure 18: Empirical real-world distribution vs. fitted Gaussian reference for Δ_i on *CiteSeer* (Degree Centrality; Red: Empirical Real, Black Dashed: Gaussian Reference $\hat{\mu} = 2.8014, \hat{\sigma} = 8.2573$, KS $D = 0.2578$).

Table 5: Comparative Statistical Summary of Local Perception Gap Δ_i under Degree Centrality (Random ER Baselines vs. Real-World Topologies).

Graph	Type	N	Mean ($\hat{\mu}$)	Std ($\hat{\sigma}$)	Skewness	Pct Paradox ($\%\Delta \geq 0$)	KS Stat (D)
Karate Club	Random (ER)	34	0.5403	1.6768	-0.1494	64.71%	0.0759
Karate Club	Real	34	14.7383	19.9787	-0.7572	85.29%	0.1203
Florentine Families	Random (ER)	14	0.7476	1.7551	0.1221	64.29%	0.1649
Florentine Families	Real	15	0.8778	1.9526	-0.2602	80.00%	0.1265
Les Miserables	Random (ER)	77	0.7209	2.3764	-0.0973	63.64%	0.0425
Les Miserables	Real	77	29.7558	47.4002	0.6213	84.42%	0.1863
Cora	Random (ER)	2641	0.9040	2.2164	0.1139	68.65%	0.0473
Cora	Real	2708	7.7337	16.6120	2.5930	84.97%	0.2466
CiteSeer	Random (ER)	3098	0.7732	1.8549	0.1847	70.63%	0.0786
CiteSeer	Real	3279	2.8014	8.2573	4.2449	81.52%	0.2578

bounded between 63.64% and 70.63%. In contrast, structural heterogeneity in real-world topologies consistently raises paradox prevalence to 80.00% – 85.29% across graph scales.

- Variance and Magnitude Expansion:** Real-world perception gaps show mean values ($\hat{\mu}$) up to 41 times larger than corresponding ER baselines (e.g., 29.7558 vs. 0.7209 in *Les Miserables*). Standard deviations ($\hat{\sigma}$) show similar order-of-magnitude expansions (expanding from 2.3764 to 47.4002), reflecting localized disparity associated with high-degree hubs.
- Heavy-Tailed Skewness vs. Symmetric Baselines:** While ER baselines display symmetric distributions centered near small positive expectations (Skewness ≈ 0), large real-world citation graphs develop strong positive skewness (Skewness = 2.5930 for *Cora* and 4.2449 for *CiteSeer*), resulting in substantially higher empirical distances ($D = 0.2466$ and $D = 0.2578$, respectively) relative to the Gaussian model.

Structural Drivers Summary : While perception gaps in Erdős–Rényi baselines represent mild, symmetric Gaussian fluctuations around small positive means, empirical networks exhibit heavy-tailed, non-Gaussian gaps driven directly by degree heterogeneity and high-degree hub nodes.

7 A New Centrality: $\exp(-\Delta_i)$

Exponential Local Centrality $\exp(-\Delta_i)$. We propose a novel local centrality metric defined for each node $i \in V$ as:

$$C_{\text{local}}(i) = \exp(-\Delta_i) \quad (27)$$

The exponential transformation maps the unbounded perception gap $\Delta_i \in (-\infty, +\infty)$ strictly onto the positive domain $(0, +\infty)$. Nodes that significantly outperform their immediate neighborhood ($\Delta_i \ll 0$) receive high scores ($C_{\text{local}} \gg 1$), indicating strong local dominance. Conversely, locally disadvantaged nodes ($\Delta_i > 0$) receive scores bounded in $(0, 1)$, while a score of $C_{\text{local}}(i) = 1$ indicates exact parity ($\Delta_i = 0$) between a node and its neighborhood average.

To evaluate whether $\exp(-\Delta_i)$ captures structural prominence comparable to standard global centralities x , we analyze the linear dependency using the Pearson correlation coefficient $r(u, v)$.

Pearson Correlation Coefficient $r(u, v)$

$$r(u, v) = \frac{\sum_{i=1}^n (u_i - \bar{u})(v_i - \bar{v})}{\sqrt{\sum_{i=1}^n (u_i - \bar{u})^2} \sqrt{\sum_{i=1}^n (v_i - \bar{v})^2}} \quad (28)$$

Intuition: Pearson correlation measures the linear alignment between two centrality vectors u and v across n nodes (where $\bar{u}$ and $\bar{v}$ denote sample means). Values near $+1$ indicate strong structural agreement, values near 0 signify linear independence, and negative values ($r < 0$) indicate an inverse relationship between global status and local dominance.

In our context, we compute $r(\exp(-\Delta), \mathbf{x}) = \text{Corr}(\exp(-\Delta_i), x_i)$ across all evaluated datasets and centrality measures $\mathbf{x}$. Empirical results are reported in Table 6 and visually depicted as a heatmap in Figure 19.

Table 6: Pearson correlation $\text{Corr}(\exp(-\Delta_i), x)$ between the proposed exponential local centrality $\exp(-\Delta_i)$ and original centrality metrics x .

Graph	Degree	Eigenvector	Katz	Exponential	Non-backtracking	Walk
Karate Club	0.5135	0.8449	0.5572	0.9405	0.8443	0.8717
Florentine Families	0.7179	0.8326	0.5491	0.8902	0.8087	0.8850
Les Misérables	0.5661	0.7332	0.4976	0.8467	0.7332	0.7731
Cora	0.6033	-0.2628	0.6607	0.5762	-0.2628	-0.1580
CiteSeer	0.4959	-0.1309	0.4820	0.3803	-0.1396	-0.0933

As reported in Table 1, the evaluated networks exhibit stark structural differences in scale and sparsity. Small social networks (*Karate Club*, *Florentine Families*, *Les Misérables*) feature high densities ($p \in [0.0868, 0.1905]$) and small node counts ($n \leq 77$), whereas citation networks (*Cora*, *CiteSeer*) are large ($n > 2700$) and sparse ($p \leq 0.0014$).

Interpretation of Degree Correlation and Local Specificity. A notable empirical observation in Table 6 is that $\exp(-\Delta_i)$ exhibits a moderate, incomplete correlation with raw degree centrality across all networks ($r \in [0.4959, 0.7179]$). This partial alignment reflects a fundamental conceptual distinction: degree measures *absolute global connectivity*, whereas $\exp(-\Delta_i)$ captures *relative local dominance*:



Figure 19: Heatmap of Pearson correlation coefficients $\text{Corr}(\exp(-\Delta_i), x)$ across evaluated centrality metrics and datasets. The color scale ranges from cool blue (-1.0 , negative alignment) to warm red ($+1.0$, strong positive alignment).

- **Local Leaders with Moderate Degree:** Nodes situated in sparse or peripheral micro-communities may possess a modest absolute degree, yet significantly outperform their immediate low-degree neighbors ($\Delta_i \ll 0$). Such nodes attain a high local dominance score $\exp(-\Delta_i) > 1$, acting as focal points within their local subgraphs despite limited global visibility.
- **High-Degree Nodes in Dense Hub Neighborhoods:** Conversely, a node with a high absolute degree that is connected to even larger global hubs may experience a positive perception gap ($\Delta_i > 0$), dampening its local score below 1.

Degree Centrality Alignment :

- The moderate correlation ($r \approx 0.50 - 0.72$) indicates that $\exp(-\Delta_i)$ does not simply duplicate degree ranking.
- Instead, it disentangles local structural leadership from absolute edge counts, accommodating peripheral leaders and contextualizing high-degree hubs.

Correlations with Global Spectral and Distance-Based Centralities. The empirical results reveal distinct behaviors across global centrality families depending on network topology:

- **Spectral Centralities (Eigenvector, Non-backtracking, Random Walk):** In small, dense social graphs, spectral metrics correlate strongly with $\exp(-\Delta_i)$ ($r \in [0.7332, 0.8717]$). In these settings, spectral weight is broadly distributed, aligning global hubs with local neighborhood leaders. However, in large, sparse graphs (*Cora*, *CiteSeer*), these correlations decorrelate or become weakly negative ($r \in [-0.2628, -0.0933]$). This trend is consistent

with *eigenvector localization*, where global spectral weight concentrates disproportionately on localized, densely connected subgraphs, uncoupling global rank from broader local neighborhood dominance.

- **Distance and Matrix-Exponential Centralities (Katz, Exponential):** Distance-decayed metrics demonstrate greater stability across topological shifts. Exponential Centrality maintains high positive correlation in dense graphs ($r \in [0.8467, 0.9405]$) and retains moderate alignment in sparse networks ($r \in [0.3803, 0.5762]$). Similarly, Katz Centrality exhibits consistent positive correlations across all datasets ($r \in [0.4820, 0.6607]$). Attenuating long-range walk weights appears to mitigate spectral localization effects, preserving a degree of alignment between local dominance and global status.

Spectral Centralities and Localization :

- Pure spectral metrics (Eigenvector, Non-backtracking, Walk) align closely with local dominance in small, dense topologies.
- In large, sparse networks, spectral localization can cause these metrics to decorrelate or negatively align with local perception gaps.

Attenuated and Distance-Based Metrics :

- Metrics with attenuation mechanisms (Katz, Exponential Centrality) maintain stable positive correlations with $\exp(-\Delta_i)$ across both dense and sparse regimes.
- Suppressing infinite-length walk weights helps preserve consistent structural agreement between local dominance and global positioning.

To visually illustrate the node-level behavior of this metric, Figure 20 visualizes the *Karate Club* network, highlighting the top 5 most locally important nodes under $\exp(-\Delta_i)$ calculated using Degree Centrality.

As shown in Figure 20, the top 5 local leaders (nodes 33, 0, 32, 2, and 1) correspond to major structural hubs in the social network. Node 33 (John A.) and Node 0 (Mr. Hi) achieve the highest scores ($\exp(-\Delta_{33}) = 1.491$, $\exp(-\Delta_0) = 1.425$) because their individual centralities far exceed the average centrality of their immediate neighbors, placing them in positions of strong local dominance.

Karate Club: Top 5 Locally Important Nodes according to $\exp(-\Delta_i)$ (Red)

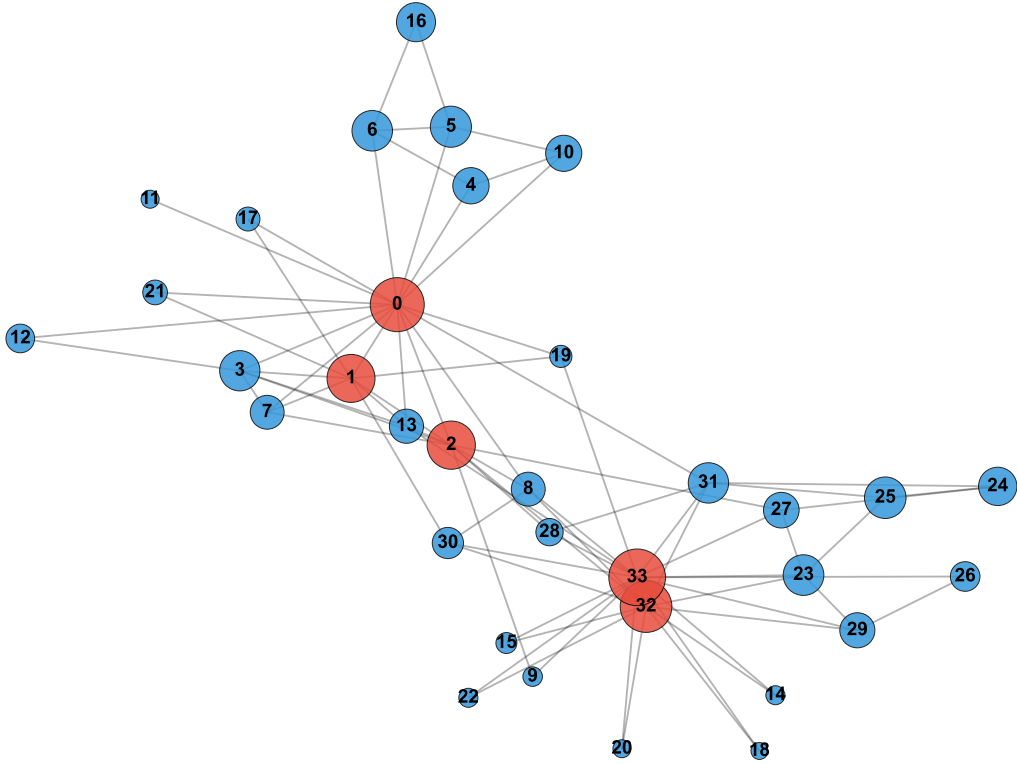


Figure 20: Visualization of the *Karate Club* graph highlighting the top 5 most locally important nodes according to $\exp(-\Delta_i)$ in red (nodes 33, 0, 32, 2, and 1). Node sizes are linearly scaled proportionally to their local score $s_i = -\Delta_i$.

8 Conclusion and Future Work

8.1 Conclusion

In this study, we presented a comprehensive structural analysis of the local perception gap Δ_i and introduced a novel exponential local centrality metric, $\exp(-\Delta_i)$, designed to capture local dominance within complex networks. By contrasting empirical observations against the homogeneous Erdős–Rényi baseline model, we demonstrated that while the friendship paradox in random graphs behaves as a minor, near-Gaussian statistical artifact governed by the Central Limit Theorem, it manifests in real-world networks as an intrinsic structural imperative. Heavy-tailed degree distributions, hub-and-spoke connectivity, and community structures systematically amplify both the prevalence (80% – 85%) and the magnitude of the perception gap, triggering a total breakdown of Gaussian assumptions and leading to extreme right-skewed distributions.

Furthermore, our correlation analysis highlighted a fundamental divergence between small, dense networks and large, sparse topologies: while $\exp(-\Delta_i)$ aligns strongly with global spectral centralities in dense graphs, localized eigenvectors in sparse citation graphs uncouple global status from local influence, positioning our proposed metric as a robust descriptor of local topological dominance.

8.2 Future Work

Building upon the empirical and analytical foundations established in this paper, several promising directions emerge for future investigation:

Extension to Alternative Random Graph Models. Beyond the Erdős–Rényi baseline, future work will examine richer topologies: scale-free Barabási–Albert networks to link preferential attachment exponents with perception gap skewness, and Watts–Strogatz small-world models to isolate how clustering and path lengths drive variance expansion in Δ_i .

Friendship Paradox Aware Graph Neural Networks. A compelling application lies in Graph Neural Networks (GNNs), where standard message-passing algorithms suffer from oversmoothing and degree bias due to high-degree hubs dominating feature aggregation. To address this, we propose integrating the local friendship paradox directly into message passing via a Friendship Paradox Aware framework:

Friendship Paradox Aware Framework.

- **Local Friendship Gap Conceptualization:** Defines a node-level metric measuring how much a local neighborhood is biased toward high-degree hubs compared to the global graph average. The effect vanishes naturally in structurally homogeneous or regular graphs.
- **Adaptive Friendship Paradox Correction:** Replaces static normalization with an adaptive, context-aware aggregation scheme. Dynamic node-specific weights increase debiasing strength for nodes surrounded by dominant hubs, dampening overrepresented feature signals during message passing.
- **Structural Invariance and Flexibility:** Automatically scales down in unbiased regions and deactivates in regular graphs. This preserves local semantic distinctions while preventing representation collapse, enabling hub-robust node classification and link prediction.

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